

Problem H2.1 [10 points]

This exercise will focus on getting the Fibonacci series using loops. The Fibonacci series is defined by taking the sum of the two previous terms of the series as follows:

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots$$

For parts (a) and (b) you should use only for-loops at any moment that you have to use a loop. Split up subquestions in different cells..

(a) Define the beginning of the Fibonacci sequence with the list `fibonacci = [1,1]`. Use a for-loop to generate the first 20 terms of the Fibonacci sequence. Store each computed number in the previously defined list. Check the length and print the resulting list. **(2pt)**

(b) Create a new list to store the 'Primonacci sequence'. Using a for-loop, loop over the previously defined Fibonacci sequence, and figure out which numbers in the sequence are prime numbers. If the numbers are a prime, store them in the new list. *Hint: To figure out if a number x is a prime, loop over the numbers and see if you find any divisors.* **(3pt)**

(c) In a new code cell, generate a list containing the first 20 terms of the Fibonacci sequence and the corresponding 'Primonacci sequence' but do use only while-loops, when you have to use a loop. Compare them with the lists generated in (a) and (b). **(4pt)**

Note: The remaining 1 point is earned by writing concise, elegant and efficient code.